Coulomb’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electric force to charge magnitude and separation.

SUCCESS CRITERIA - I can...

1. Identify charge signs before predicting direction.
2. Use inverse-square scaling.
3. Calculate force with charge in coulombs and distance in meters.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “charge” and “separation” to explain your model or result. Sentence starters are optional.

charge: a property responsible for electric interactions

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A shallow-water model uses $v=\sqrt{(gh)}$. With g=10, h=30 m and travel distance 5 km, find speed and travel time. **State** a model limitation.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = k|q_1q_2|/r^2$; $k = 9.0\times10^9\ N\cdot m^2/C^2$.
r is separation. Signs determine attraction/repulsion; the equation gives magnitude.

Worked reasoning: Doubling the separation between two fixed charges makes Coulomb-force magnitude one quarter as large. The charge signs determine attraction or repulsion separately.

Your turn: Sketch both separations for two positive charges and label the force directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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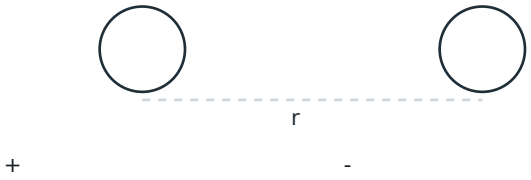
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two opposite charges are moved twice as far apart.
Draw their force directions and compare the new force with the original.

Sentence starter: Both … . They differ in … because … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

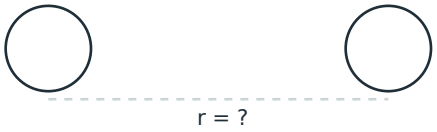
Marzano check - circle: 1 2 3

04 Apply the physics

Two small charged spheres carry +2.0 μC and -3.0 μC and are 0.30 m apart.
Use $k = 9.0 \times 10^9\ N \cdot m^2/C^2$ and $1\ \mu C = 10^{-6}\ C$.
Label the model, calculate force magnitude and draw the force on each sphere.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Charge 1 Charge 2

Marzano check - circle: 1 2 3

05 Explain your reasoning

One charge has twice the magnitude of the other. Does it exert twice the force on the smaller charge?
Explain.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 600 Hz tone lasts 0.4 s.
Find number of cycles and period.
Label a wave sketch to support the calculation or comparison.

Sentence starter: I found … [unit]. This value means … in this situation.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Both charge magnitudes double and distance triples. Express the new force as a fraction of the old.

Sentence starter: I notice … . This supports … because … .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two small spheres carry +1.0 μC and +4.0 μC and are 0.20 m apart.
Use $k = 9.0 \times 10^9\ N \cdot m^2/C^2$.
Calculate the force magnitude and draw both force arrows. Then triple the separation and recalculate.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3